ИТМО

Analysis & Development of Algorithms

VI

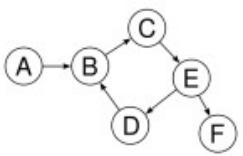
Anton Kovantsev
https://t.me/pop_antonij

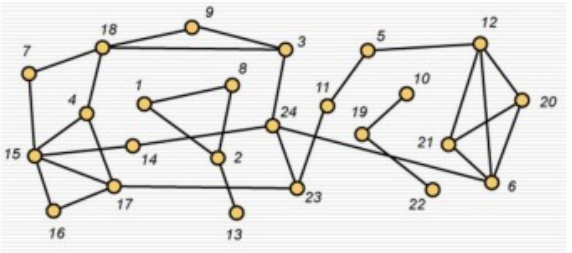
Problem statement

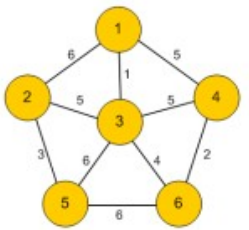
How to distinguish networks (graphs)?
Which concepts and measures should be used for network analysis?

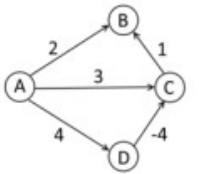
Recall the definition of:

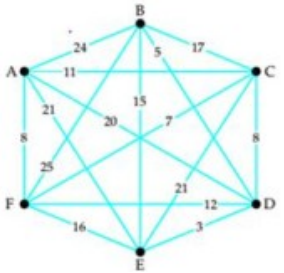
- directed and undirected graphs;
- weighted and unweighted graphs;
- complete and non-complete graphs;
- connected and disconnected graphs;
- dense and sparse graphs.

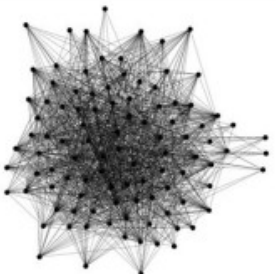


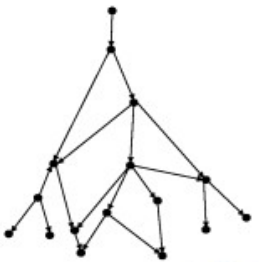














Bank group subscribers (BGS) network

A network of VK bank group subscribers represented by an unweighted undirected connected graph with 15,923 nodes and 200,633 edges.

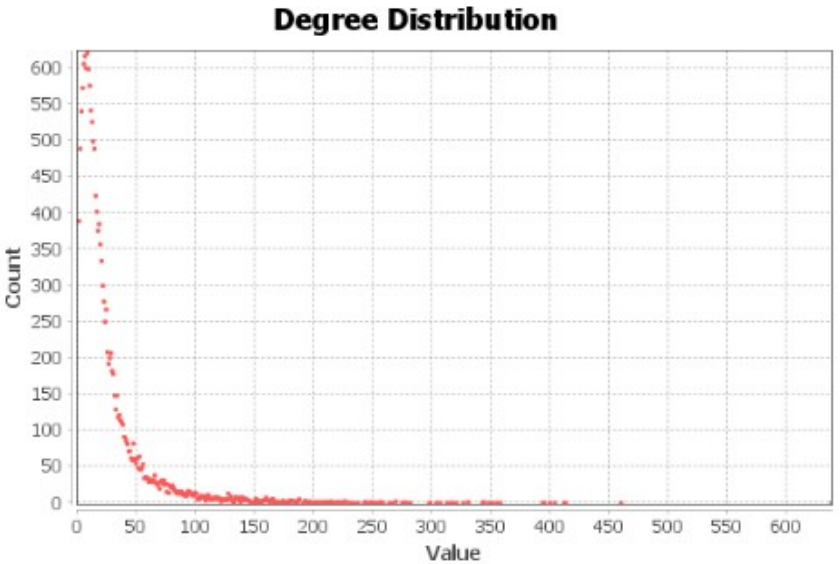
How can we handle (analyse, use) this network?



Basic and degree measures

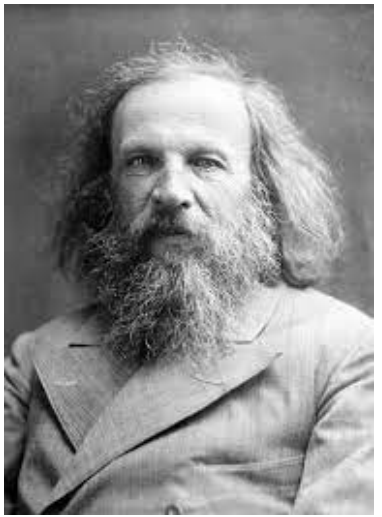
Basic measures:
|V| - the number of vertices; |E| - the number of edges
Degree measures:

$d(v)$, degree of v, i.e. the number of edges for vertex v
$d_{\text{in}}(v)$, in-degree of v, i.e. the number of in-edges for vertex v
$d_{\text{out}}(v)$, out-degree of v, i.e. the number of out-edges for vertex v
$\bar{d} = \frac{1}{|V|} \sum_{v \in V} d(v)$, average degree over all vertices



Values for BGS:
|V| = 15, 923
|E| = 200, 633
$d_{average}$ = 25.20
Large part: low d
Small part (aka hubs): high d

Statistical hypotheses about the distribution?



Наука начинается с тех пор, как начинают измерять. Точная наука немыслима без меры. (*Science begins when we begin to measure. One can't imagine exact science without measure*).

Scale-free network

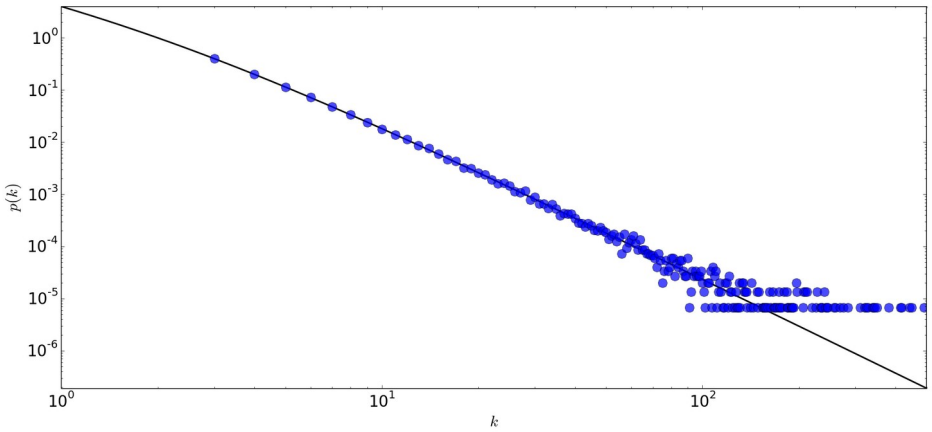
A scale-free network is a network whose degree distribution follows a power law, at least asymptotically. That is, the fraction P(k) of nodes in the network having k connections to other nodes goes for large values of k as

$$P(k) \sim k^{-\gamma}; 2 < \gamma < 3$$

The name "scale-free" could be explained by the fact that some moments of the degree distribution are not defined, so that the network does not have a characteristic scale or "size".

The name "scale-free" could be explained by the fact that some moments of the degree distribution are not defined, so that the network does not have a characteristic scale or "size".

Albert-László Barabási and Réka Albert proposed a generative mechanism to explain the appearance of power-law distributions, which they called "preferential attachment"



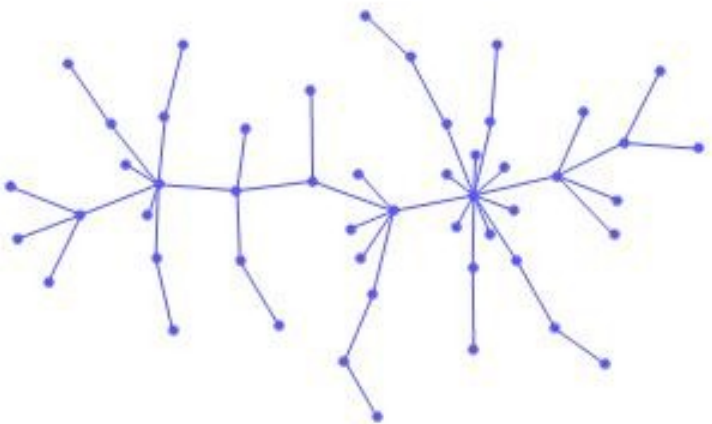
The **Barabási–Albert (BA) model** is an algorithm for generating random scale-free networks using a preferential attachment mechanism. Several natural and human-made systems, including the Internet, the World Wide Web, citation networks, and some social networks are thought to be approximately scale-free and certainly contain few nodes (called hubs) with unusually high degree as compared to the other nodes of the network.

Barabási and Albert model

The only parameter in the BA model is m, a positive integer.
The network initializes with a network of $m_0 \geq m$ nodes.
At each step, add one new node, then sample m existing vertices from the network, with a probability that is proportional to the number of links that the existing nodes already have:

$$P_i = \frac{k_i}{\sum_j k_j} \text{ where } k_i \text{ is the degree of node } i \text{ and the sum is made over all pre-existing nodes } j$$

Heavily linked nodes ("hubs") tend to quickly accumulate even more links, while nodes with only a few links are unlikely to be chosen as the destination for a new link. The new nodes have a "preference" to attach themselves to the already heavily linked nodes.



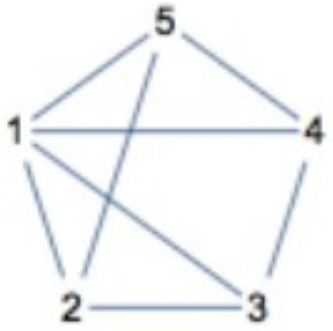
A tree network generated according to the Barabasi-Albert model. The network is made of 50 vertices with initial degrees $m_0 = 1$

Matthew principle, is the tendency of individuals to accrue social or economic success in proportion to their initial level of popularity, friends, and wealth. It is sometimes summarized by the adage or platitude *"the rich get richer and the poor get poorer"* (Math. XXV 29)

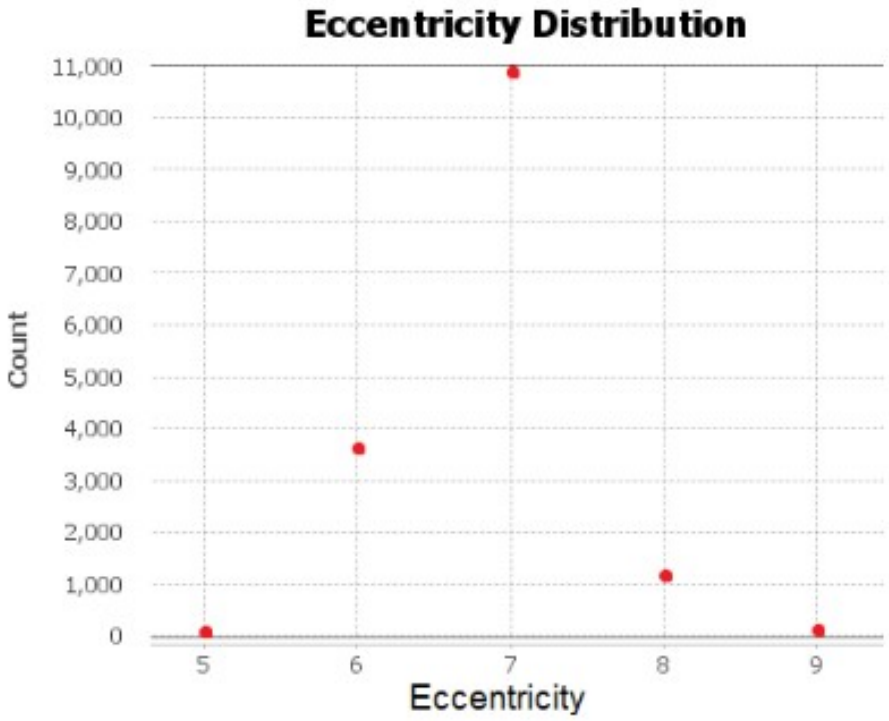
Distance measures

Given a connected G, $dist(v,u)$ is the distance (shortest path length) between v and u
The eccentricity $\epsilon(v)$ of v is the greatest distance between v and any other vertex:
$\epsilon(v)=\max_{u\in V} dist(v,u)$ ("how far a node is from the node most distant from it").
The radius r is the minimum eccentricity of any vertex:
$r=\min_{v\in V}\epsilon(v)=\min_{v\in V}\max_{v\in V} dist(v,u)$.
The diameter D is the maximum eccentricity of any vertex, i.e.
the greatest distance between any pair of vertices: $D=\max_{v\in V}\epsilon(v)$.

The average path length $l=\frac{1}{|V|(|V|-1)}\sum_{u\neq v} dist(v,u)$
("the efficiency of information or mass transport on a network").



Values for BGS:
$r=5$
$D=9$
$l=3.48$



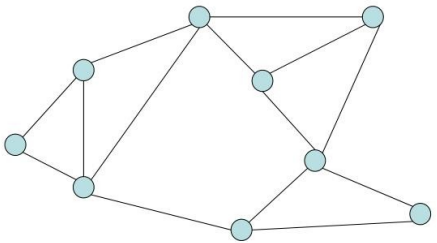
Connectivity

A graph G (complete or not) is said to be k-vertex-connected if it contains at least k + 1 vertices, but does not contain a set of k − 1 vertices whose removal disconnects the graph; and κ(G) is defined as the largest k such that G is k-connected.

The edge-connectivity λ(G) is the size of a smallest edge cut, and the local edge-connectivity λ(u, v) of two vertices u, v is the size of a smallest edge cut disconnecting u from v.

k-Connected Graph

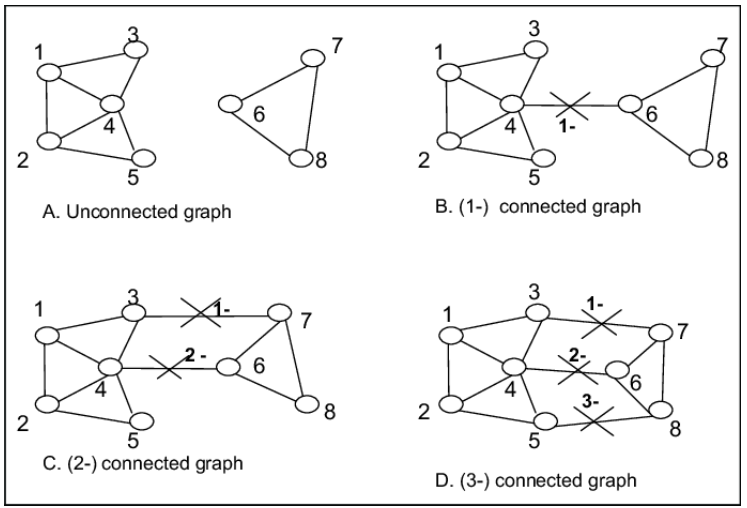
- **k-Connected Graph:** The graph whose connectivity is at least k.



κ(G)=2

G is a 2-connected graph

Is G a 1-connected graph?

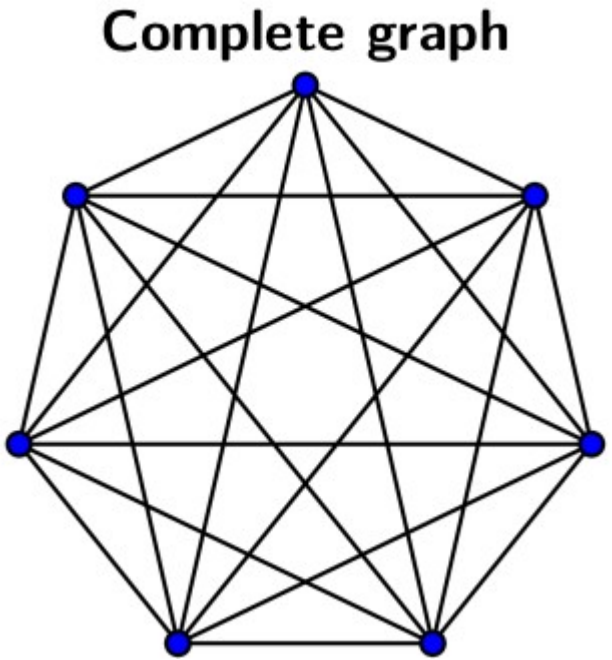


The vertex-connectivity of a graph is less than or equal to its edge-connectivity. That is, κ(G) ≤ λ(G).

Density measures

The density ρ of an undirected G is the ratio of |E| and the number of possible edges, i.e. the number of edges in the complete graph with the same |V|:

$$\rho = \frac{|2E|}{|V|(|V|-1)} \quad if\ \rho \approx 0 \Rightarrow sparse\ \ graph$$



Values for BGS:

$|V| = 15{,}923$

$|E| = 200{,}633$

$\rho = 0.002$

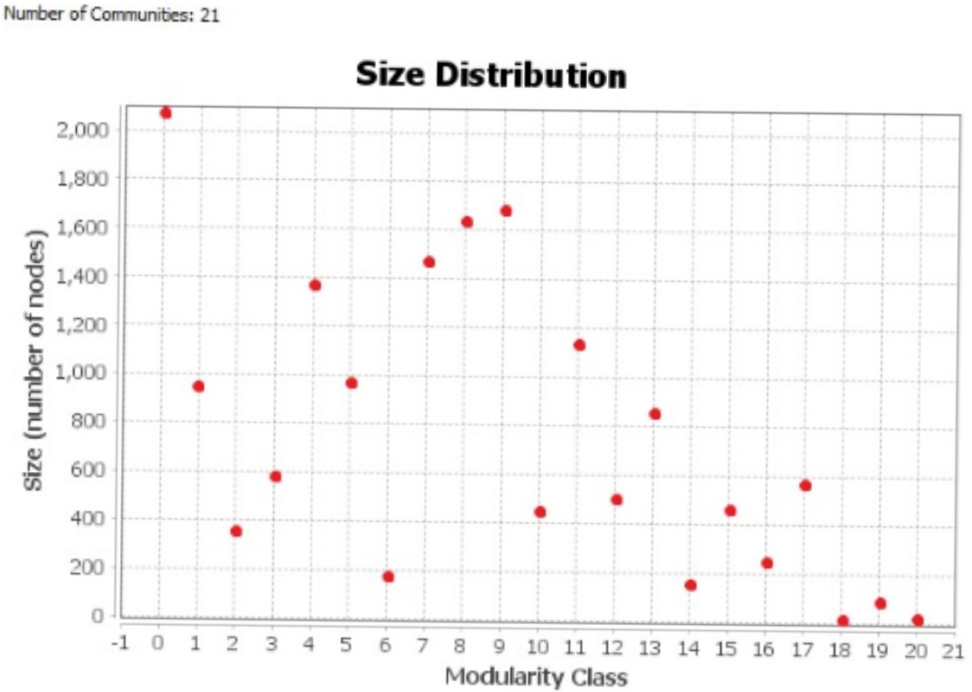
$$|E| = \frac{|V|(|V|-1)}{2}$$

$$if\ |V| = 15{,}923\ , then |E| = 126{,}763{,}003$$

Modularity

Modularity Q measures the strength of division of a graph into clusters (subgraphs, modules). Graphs with high Q > 0 have dense connections between the vertices within clusters but sparse between those in different clusters.
Q compares the number of edges within clusters in G with **the expected number of edges in a random graph** regardless of clusters.

Values for BGS: Q = 0.463



Random distribution of edges between all vertices

Configuration model (CM) is a family of random graph models designed to generate networks from a given degree sequence. For G with v having d(v), CM cuts each edge into halves (each called a stub), and then each stub is rewired randomly with any other stub in G (except itself). Thus, d(v)-distribution remains the same but CM results in a new random $\tilde{G}$. It provides the benchmark in the calculation of network modularity.

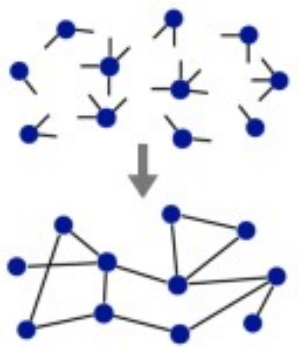
The expected number of edges between v , u $\in$ $\tilde{G}$

The total number of stubs is $\sum_{w \in V} d(w) = 2|E|$

For i = 1, . . . , d(v), let s_i = 1 if the i-th stub of v connects to one of stubs of u and s_i = 0, otherwise.
Since the i-th stub of v can connect to any of the 2|E| − 1 remaining stubs with equal probability and since there are d(u) stubs of u, $\mathrm{E}[s_i] = \frac{d(u)}{2|E|-1}$

The total number of edges between v and u is $J_{u,v} = \sum_{i=1}^{d(v)} s_i$ so,

$$\mathrm{E}[J_{u,v}] = \sum_{i=1}^{d(v)} \mathrm{E}[s_i] = \frac{d(u)d(v)}{2|E|-1} \approx \frac{d(u)d(v)}{2|E|}$$



Calculating Modularity Q

The difference between the actual number A_{vu} of edges between v and u (from the adjacency matrix A) and the expected number of edges between them is:

$$\Delta(u,v) = A_{v,u} - \frac{d(v)d(u)}{2|E|}$$

Represents the expected probability of an edge between nodes u and v under the configuration model.

Let C be a division of G into clusters and c(v) denote the cluster of v.
If c(v) = c(u), Q should increase if Δ(u, v) > 0 and decrease otherwise.
If c(v) ≠ c(u), Q should not change.
Thus, after normalization,

$$Q(C) = \frac{1}{2|E|} \sum_{u,v \in V} \left(A_{v,u} - \frac{d(v)d(u)}{2|E|} \right) \frac{s_u s_v + 1}{2} \qquad s_u = \begin{cases} 1 & u \in C_1 \\ -1 & u \in C_2 \end{cases}$$

$$Q = max_C Q(C)$$

Good for partitioning into two communities only.Hierarchical partitioning is a possible approach to identify multiple communities in a network.

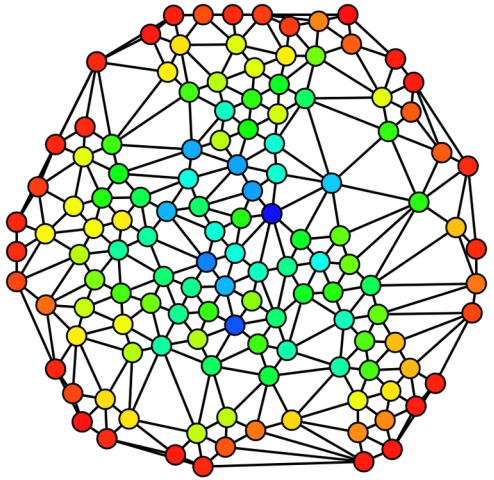
To calculate Q, one has to find divisions C of G providing maximal Q(C). It can be approximately done via numerical optimization (Fast Greedy, Louvain, etc.).

Newman and Girvan, 2004; Newman, 2006

The Girvan–Newman algorithm

The Girvan–Newman algorithm (named after Michelle Girvan and Mark Newman) is a hierarchical method used to detect communities in complex systems.

Vertex betweenness is an indicator of highly central nodes in networks. For any node v, vertex betweenness is defined as the fraction of shortest paths between pairs of nodes that run through it. It is relevant to models where the network modulates transfer of goods between known start and end points, under the assumption that such transfer seeks the shortest available route.
"Edge betweenness" of an edge as the number of shortest paths between pairs of nodes that run along it.



A directed graph colored based on the betweenness centrality of each vertex from least (red) to greatest (blue).

The betweenness centrality of a node v

$$g(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}};$$ σ_{st} - total number of shortest paths from node s to node t;

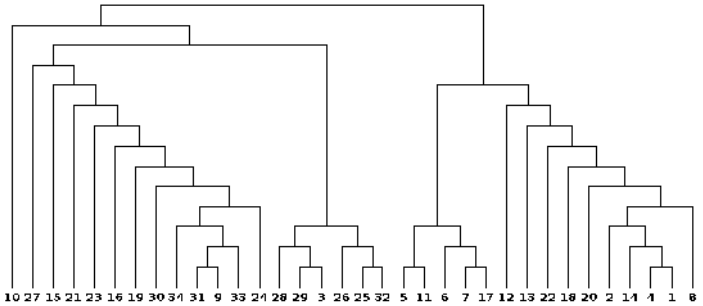
$\sigma_{st}(v)$ those paths that pass through v

The algorithm's steps for community detection are summarized below

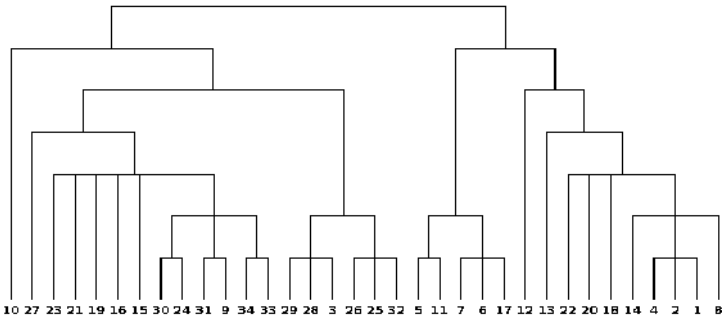
1) The betweenness of all existing **edges** in the network is calculated first.
2) The edge(s) with the highest betweenness are removed.
3) The betweenness of all edges affected by the removal is recalculated.
4) Steps 2 and 3 are repeated until no edges remain.

The Girvan–Newman algorithm

TThe end result of the Girvan–Newman algorithm is a dendrogram. As the Girvan–Newman algorithm runs, the dendrogram is produced from the top down (i.e. the network splits up into different communities with the successive removal of links). The leaves of the dendrogram are individual nodes.



(a) Single edge removal



(b) Multiple edge removal

For each resulting graph state (where the network is partitioned into (2, 3, 4, ... N) communities), calculate the modularity value (Q). Sometimes, the modularity plot may exhibit multiple peaks (for example, a local maximum at (k=2) and a global maximum at (k=5)). This indicates a multi-level (hierarchical) structure within your network. In such cases, the choice depends on your specific objective: whether you require a coarse-grained partition or a finer level of detail.

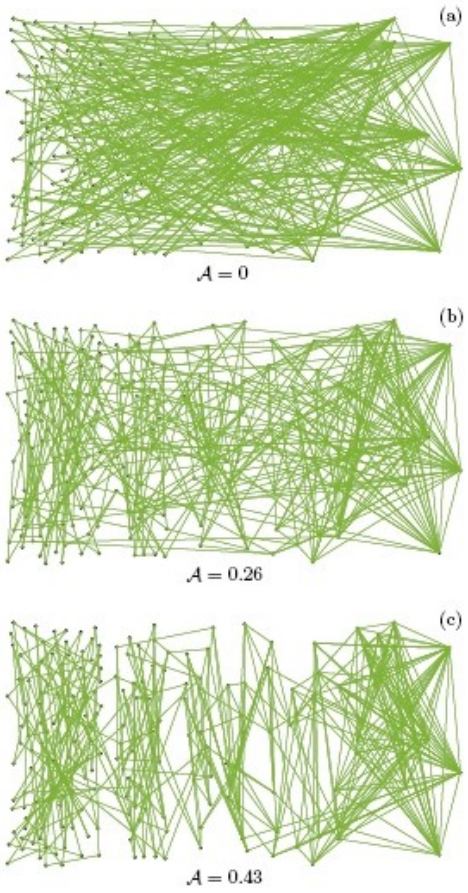
Assortativity

Similis simili gaudet

Assortativity, or assortative mixing, is a preference for a network's nodes to attach to others that are similar in some way. Though the specific measure of similarity may vary, network theorists often examine assortativity in terms of a node's degree. The addition of this characteristic to network models more closely approximates the behaviours of many real world networks.
Assortativity is often operationalized as a correlation between two nodes.
The **assortativity coefficient** is the Pearson correlation coefficient of degree between pairs of linked nodes.

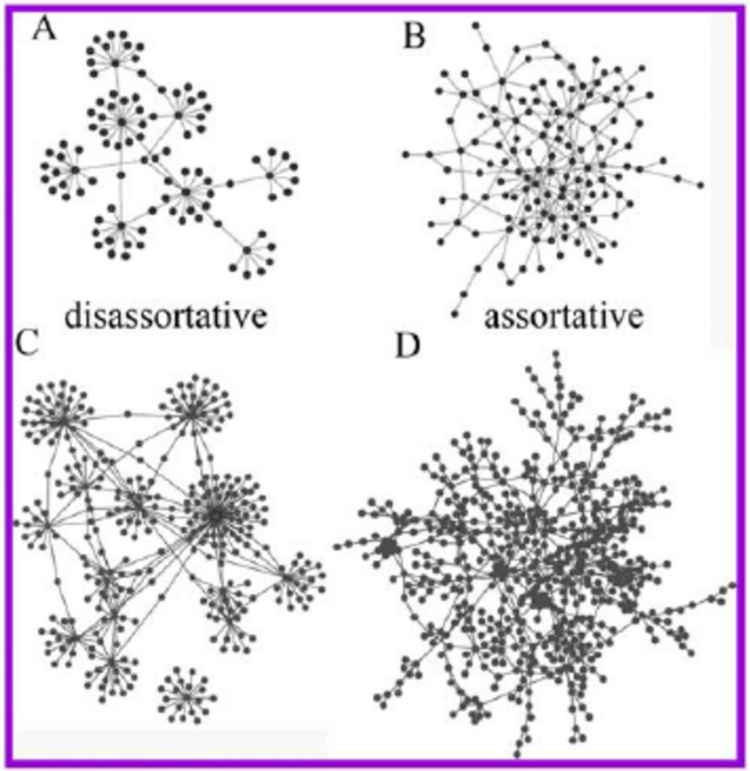
$$r = \frac{\sum_{j,k} jk\left(e_{j,k} - q_j q_k\right)}{\sigma_q^2}; \quad q_k = \frac{(k+1)\,p_{k+1}}{\sum_{j>1} jp_j}$$

q_k is the distribution of the remaining degree derived from the degree distribution p_k
$e_{j,k}$ refers to the joint probability distribution of the remaining degrees of the two vertices .
Joint probability distribution is the corresponding probability distribution on all possible pairs of outputs.



Assortativity

• **Assortative mixing (r > 0):** nodes with a high number of connections tend to connect to other high-degree nodes, while low-degree nodes connect to other low-degree nodes. This is characteristic of social networks (people are more likely to be friends with others who have a similar number of acquaintances).
• **Disassortative mixing (r < 0):** high-degree nodes connect to low-degree nodes (forming "stars" or centralized hubs). This is characteristic of technological and biological networks (e.g., internet routers or protein interactions).
• **Non-assortative mixing (r ≈ 0)**: connections are distributed randomly (as in classical Erdős–Rényi models).

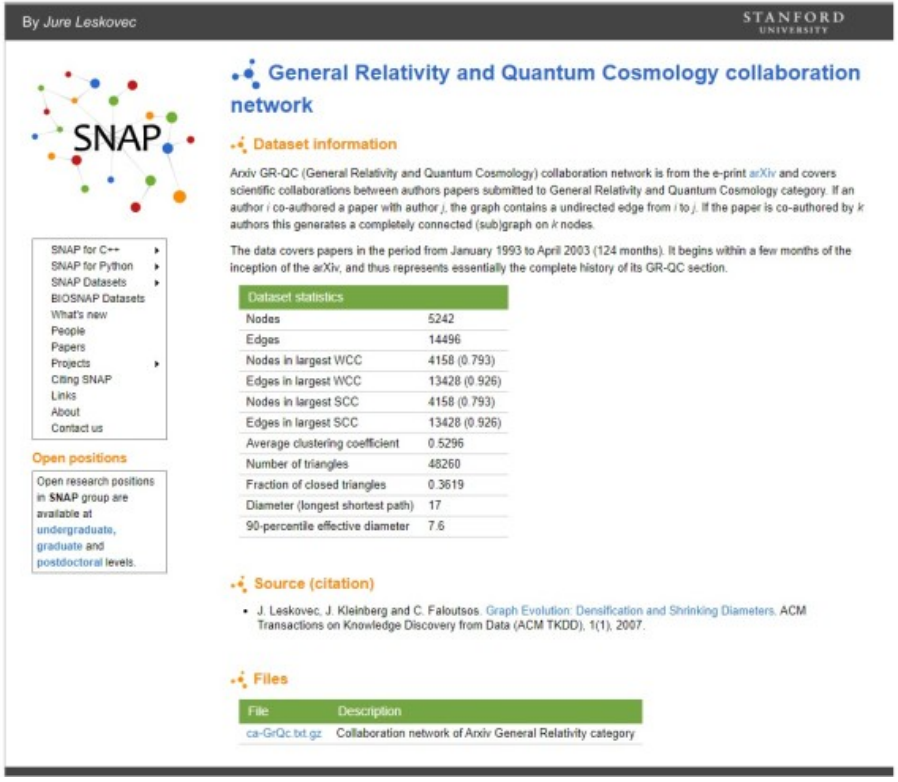


Assortativity coefficient (r)
To measure this property, Pearson's assortativity coefficient is used; it ranges from --1 to 1:
• r = 1 — perfect assortativity.
• r = 0 — no correlation (random graph).
• r = -1 — complete disassortativity.

Network analysis with Gephi

“**Gephi** is the leading visualization and exploration software for all kinds of graphs and networks. Gephi is open-source and free.” — https://gephi.org/ (seems, forbidden in RF)

Choose a graph from https://snap.stanford.edu/data/ ; Change the format for that accepted by Gephi, if necessary





Антон Кованцев
pop.antonij@gmail.com
https://t.me/pop_antonij